QUALITY INFORMATION DOCUMENT

For OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_bottom_salinity_arkona_bornholm

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko J. Linders	P. Lagemaa
1.1	2019/02/21		Revision	U. Raudsepp I. Maljutenko	P. Lagemaa
2.0	2019/10/10		Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Remove temporal extension specific details	U. Raudsepp I. Maljutenko	P. Lagemaa
3.0	2023/09/01	all	Template change+ minor changes	U. Raudsepp I. Maljutenko	P. Lagemaa
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1) INTRODUCTION

a) Product Name

OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_bottom_salinity_arkona_bornholm.

b) Indicator Evaluation Method

This document provides to Copernicus Marine Service users information on the quality of the bottom salinity in the Arkona and Bornholm Basins of the Baltic Sea. These sites were chosen to represent the inflow events of the Baltic Sea and therefore grouped as a separate CMEMS product which is named BALTIC_OMI_WMHE_mbi_bottom_salinity_arkona_bornholm.

The bottom salinity data in the Arkona and Bornholm Basins are daily values extracted from the BALTICSEA_MULTIYEAR_PHY_003_011 product processed with 7-day moving average filter.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf

2) SCIENTIFIC VALIDATION

a) Major Baltic Inflow: bottom salinity

The accuracy of the index is directly dependent on the accuracy of the Baltic reanalysis product (BALTICSEA_MULTIYEAR_PHY_003_011).

The model bottom salinity product is validated against long term time series measurements. At the Arkona Basin, the mean bias of the bottom salinity and cRMSD are -0.01 and 1.95 PSU, respectively. At the monitoring station BMPK2 which is the closest to the Bornholm Basin, the bottom salinity bias and cRMSD are -0.01 and 0.45 PSU, respectively. Direct comparison with in-situ observations are presented in Figure 1.

Scientific evaluation can be found in the Copernicus Marine Service Ocean State Report (Raudsepp et al., 2016; Raudsepp et al., 2018).

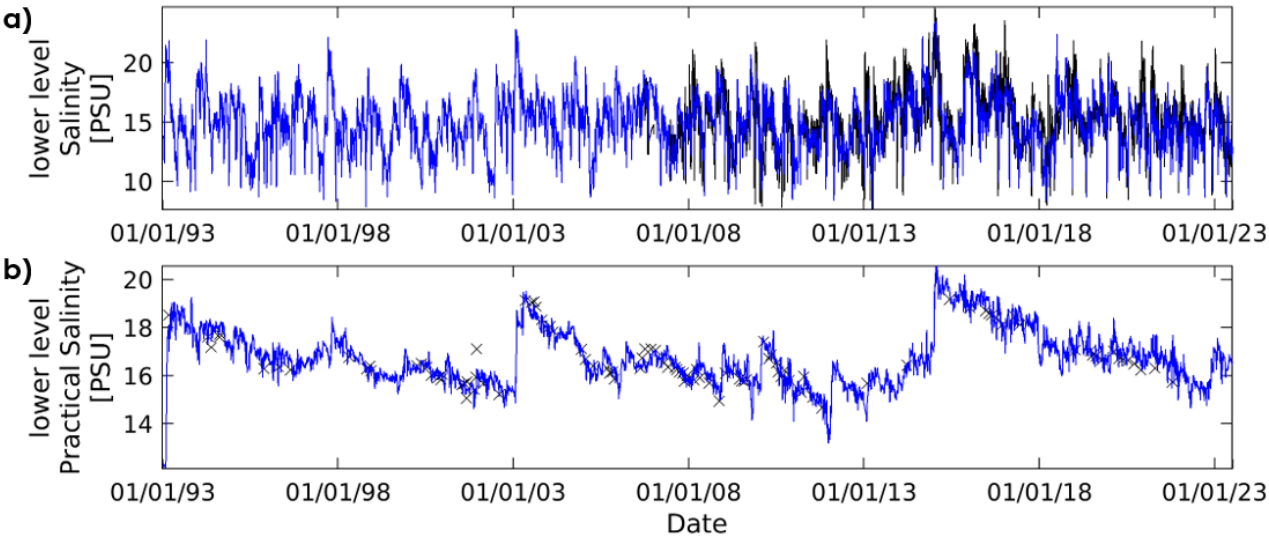

Figure 1. Direct comparison of modelled bottom salinity (blue line) with in-situ observations (black marker): a) Arkona, b) Bornholm.

3) REFERENCES

Raudsepp et al., 2016: Major Baltic inflows. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

Raudsepp, U., Legeais, J.-F., She, J., Maljutenko, I., Jandt, S., 2018. Baltic inflows. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, s106–s110, DOI: 10.1080/1755876X.2018.1489208